Carlos Abarca

Antioch, CA

(510) 550-6204 | cabarca@ucsb.edu | Portfolio | LinkedIn | Handshake

Education

University of California, Santa Barbara — **Aug. 2023 – Sept. 2025**
Bachelor of Science in Mechanical Engineering — *Santa Barbara, CA*

Diablo Valley College — **Aug. 2020 – Jun. 2023**
Pre-Engineering Coursework — Mechanical Engineering Transfer — *Pleasant Hill, CA*

Professional Experience

Penumbra Inc. — **Jun. 2026 – Present**
Equipment Technician 1 — *Alameda, CA*

- Performed preventative maintenance and repairs on production equipment per documented procedures.
- Collaborated with engineers to troubleshoot equipment using root cause analysis.
- Built fixtures per engineering drawings and specifications.
- Maintained documentation in compliance with medical device regulations; worked in cleanroom environments.

Academic Experience

True Digital Surgery — Capstone Sponsor — **Aug. 2024 – Jun. 2025**
Mechanical Engineering Student — *Santa Barbara, CA*

- Collaborated with a 5-member team to resolve thermal issues in a surgical camera by relocating LEDs to the base and routing light through four fiber-optic bundles, improving reliability.
- Authored SOPs for inspection, verification, and thermal testing of PCB/LED assemblies on aluminum heat sinks; validated optical output across white, IR, UV, and fluorescence bands.
- Performed soldering, rework, and troubleshooting of prototype PCBs.
- Designed aluminum light-module housing and optical collimator in SolidWorks.
- Produced machinable GD&T drawings and collaborated with machinists to improve DFM, reduce cost, and integrate off-the-shelf components.
- Led rapid prototyping of 3D-printed FFF models for tolerance checks, transitioning to CNC-machined 6061-T6 aluminum for final integration.
- Conducted coupled SolidWorks–COMSOL FEA and thermal analyses for mass minimization while maintaining a 2× safety factor.

Personal Projects

Evasive Robotic Toy for Cats | *Arduino, HC-SR04, Motor Control* — **Jan. 2026 – Jun. 2026**

- Prototyped and tested motor and sensor options, selecting final hardware based on performance requirements.
- Iterated on initial design and migrated from breadboard wiring to a custom 2-layer PCB regulating 11.1V LiPo input down to 6V (motor) and 5V (logic/sensor) rails via dual buck converters, optimizing power/signal routing.
- Designed a mouse-shaped chassis in Fusion 360, hand-calculating clearances to fit motors, caster, sensors, battery mounts, and PCB within a compact shell; 3D-printed multiple iterations to validate fit.
- Coded obstacle-avoidance and evasion logic in Arduino, integrating sonar sensing with real-time motor control for reactive maneuvering.

Bike Wheel ABS Prototype | *Arduino, PID Control, Mechanical Design* — **Jun. 2024**

- Designed and prototyped a bicycle anti-lock braking (ABS) system integrating a DC motor, cam-follower brake tensioner, encoder feedback, and IR sensing; implemented Arduino-based PID control to detect rapid deceleration and dynamically modulate brake tension, reducing wheel lock-up.
- Built modular hardware including 3D-printed housings, custom sensor mounts, and dual limit switches for tolerance verification, reliable actuation, and repeatable testing during iterative development.

Technical Skills

CAD & Software: SolidWorks, COMSOL Multiphysics, Fusion 360, MATLAB, Arduino, KiCad, LabVIEW
Analysis: FEA, Thermal Simulation, Parametric & Generative Design, PID Control
Tools: 3D Printing, Oscilloscope, Soldering
Languages: English (native), Spanish (native)

Awards & Honors

Salutatorian — **Jun. 2020**
Pittsburg High School (Class Rank #2/820) — *Pittsburg, CA*